Exponential Distribution

1 Definition

A positive continuous random variable X is said to have an Exponential distribution if its PDF is given by,

$$f_X(x) = \begin{cases} \frac{1}{\theta}e^{-x/\theta}, & \text{if } 0 < x < \infty \\ 0, & \text{otherwise} \end{cases} \quad \text{where} \theta > 0$$

Notation: $X \sim \text{Exp}(\theta) \Leftrightarrow X \sim \text{Gamma}(\theta, n = 1)$

2 CDF

$$F_X(x) = \int_{-\infty}^{x} f_X(t)dt = \begin{cases} 0 & x \leqslant 0 \\ \int_0^x \frac{1}{\theta}e^{-t/\theta}dt & x > 0. \end{cases}$$

$$\begin{aligned} &= \int_0^x \frac{1}{\theta}e^{-t/\theta}dt \\ &= \frac{1}{\theta}\left[\frac{e^{-t/\theta}}{-1/\theta}\right]_0^x \\ &= -\left[e^{-x/\theta} - e^{-\theta}\right] \\ &= 1 - e^{-x/\theta} \end{aligned}$$

Clearly, $x = 0$ is the mode and it is an extremely positively skewed dist.

3 Median

For a continuous distribution,

$$\begin{aligned} & F_X(me) = 1/2 \\ \Rightarrow & 1 - e^{-me/\theta} = 1/2 \\ \Rightarrow & e^{-me/\theta} = 1/2 \\ & \Rightarrow -me/\theta = \log(1/2) \\ & \Rightarrow me = \theta \log 2 \end{aligned}$$

4 Moments

$$\begin{aligned} E\left(X^n\right) &= \int_0^\infty x^n \frac{1}{\theta}e^{-x/\theta}dx \\ &= \frac{1}{\theta}\theta^{n+1}\Gamma(n+1) = \theta^n\Gamma(n+1) \end{aligned}$$

If n is an integer, $E\left(X^n\right) = \theta^n n!$

$$\begin{aligned} &E(X) = \theta \qquad \text{Var}(X) = E\left(X^2\right) - E^2(X) \\ &E\left(X^2\right) = \theta^2 2 \qquad\quad = 2\theta^2 - \theta^2 = \theta^2 \end{aligned}$$

5 Mean deviation about median

$$
\begin{aligned}
E|X-me| &= E|X-\theta\ln 2| \\
&= \int_0^{\theta\ln 2}(\theta\ln 2-x)\frac{1}{\theta}e^{-x/\theta}dx + \int_{\theta\ln 2}^{\infty}(x-\theta\ln 2)\frac{1}{\theta}e^{-x/\theta}dx \\
\text{Now, } \int_0^{\theta\ln 2}(\theta\ln 2-x)\frac{1}{\theta}e^{-x/\theta}dx &= \frac{1}{\theta}\int_{\theta\ln 2}^{0} ze^{-(\frac{\theta\ln 2-z}{\theta})} - dz, \quad \theta\ln 2 - x = z, \quad -dx = dz \\
&= \frac{1}{\theta}\int_0^{\theta\ln 2} ze^{-(\frac{\theta\ln 2-z}{\theta})}dz \\
&= \frac{1}{\theta}e^{-\ln 2}\int_0^{\theta\ln 2} ze^{z/\theta}dz \\
&= \frac{1}{\theta}e^{\ln z^{-1}}\left[z\frac{e^{z/\theta}}{1/\theta}\bigg|_0^{\theta\ln 2} - \int_0^{\theta\ln 2}\frac{e^{2/\theta}}{1/\theta}dz\right] \\
&= \frac{1}{2\theta}\left[\theta\cdot\theta\ln 2.2 - \theta\frac{e^{z/\theta}}{1/\theta}\bigg|_0^{\theta\ln 2}\right] \\
&= \frac{1}{2\theta}\left[2\theta^2\ln 2 - \theta^2(2-1)\right]. \\
&= \frac{1}{2\theta}\left(\theta^2 2\ln 2 - \theta^2\right) \\
&= \theta\ln 2 - \frac{\theta}{2} \\
\int_{\theta\ln 2}^{\infty}(x-\theta\ln 2)\frac{1}{\theta}e^{-x/\theta}dx &= \frac{1}{\theta}\int_0^{\infty} ze^{-(\frac{z+\theta\ln 2}{\theta})}dz \\
&= e^{-\ln 2}\frac{1}{\theta}\int_0^{\infty} z^{2-1}e^{-z/\theta}dz \\
&= \frac{1}{2}\frac{1}{\theta}\frac{\Gamma 2}{\left(\frac{1}{\theta}\right)^2} \\
&= \frac{1}{2\theta}\theta^2 = \theta/2 \\
\therefore E(X-me) &= \theta\ln 2 - \frac{\theta}{2} + \frac{\theta}{2} \\
&= \theta\ln 2
\end{aligned}
$$

1. If X has an exponential distribution with mean θ, show that $\dfrac{\text{Mean deviation about mean}}{\text{SD}} = \dfrac{2}{e}$.

$$
\begin{aligned}
\text{Mean deviation about mean} &= E|X-\theta| \\
&= 2\int_{\theta}^{\infty}(x-\theta)\frac{1}{\theta}e^{-x/\theta}dx, \quad x\theta = z \Rightarrow dx = dz \\
&= \frac{2}{\theta}\int_0^{\infty} ze^{-\frac{z+\theta}{\theta}}dz \\
&= \frac{2}{\theta e}\int_0^{\infty} ze^{-z/\theta}dz \\
&= \frac{2}{\theta e}\int_0^{\infty} z^{2-1}e^{-2/\theta}dz \\
&= \frac{2}{\theta e}\theta^2\sqrt{2} = \theta\frac{2}{e}
\end{aligned}
$$

$\therefore \dfrac{\text{Mean deviation about mean}}{\text{SD}} = \dfrac{2}{e}$

Mean deviation about mean can't exceed standard deviation as $2 < e$.

6 Lack of Memory Property

If X has an exponential distribution then $P(X > a+b \mid X > a) = P(X > b)\ \forall a, b > 0$.

Proof: Let the PDF of X be $f_X(x) = \begin{cases} \frac{1}{\theta}e^{-x/\theta} & x > 0 \\ 0 & \text{otherwise} \end{cases}$

$$\therefore P(X > k) = 1 - P(X \leqslant k) = 1 - \left(1 - e^{-k/\theta}\right) = e^{-k/\theta}, \quad k > 0$$

$$P[X > a+b \mid X > a] = \frac{P[X > a+b]}{P[X > a]} = \frac{e^{-(a+b)/\theta}}{e^{-a/\theta}} = e^{-b/\theta} = P(X > b)$$

Interpretation: If X represents the life time of a given object say an elective bulb. The above theorem states that the probability that the bulb will be operative for at least $(a+b)$ units given that it has already run for a unit, is same as the initial probability of lasting for b units.

This means that the failure lifetime of an individual has the some distribution, no matter how old it is at present. This is the lack of memory property of Exponential distribution. Another way to saying is that an old functioning component has the same life time distribution as a new functioning component or the compact is not subject to fatigue.

2. Let $F(t)$ be the probability that a system fails by time t and let $\delta(t)\Delta t + O(\Delta t)$ be the probability of failure in the interval $(t, t+\Delta t)$. Given that it has already survived upto t, show that $f(t)$ satisfies following differential equation.

$$\frac{F'(t)}{1 - F(t)} = \delta(t) \quad 0 < t < \infty$$

and $F(t) = 1 - \exp\left[1 - \int_0^t \delta(x)dx\right]$

Let T be the life time of a system then $P[T \leqslant t] = P(t)$ is the DF of T.

$$\begin{aligned}
&P[t < T < t + \Delta t \mid T > t] = \delta(t)\Delta t + O(\Delta t) \\
\Rightarrow &\frac{P[t < T < t + \Delta t, T > t]}{P[T > t]} = \delta(t)\Delta t + O(\Delta t) \\
\Rightarrow &\frac{P[t < T < t + \Delta t]}{P[T > t]} = \delta(t)\Delta t + O(\Delta t) \\
\Rightarrow &\frac{\frac{F(t+\Delta t) - F(t)}{\Delta t}}{1 - F(t)} = \delta(t) + \frac{O(\Delta t)}{\Delta t} \\
\Rightarrow &\lim_{\Delta t \to 0}\left(\frac{\frac{F(t+\Delta t) - F(t)}{\Delta t}}{1 - F(t)}\right) = \delta(t) + \lim_{\Delta t \to 0}\frac{O(\Delta t)}{\Delta t} \\
\Rightarrow &\frac{F'(t)}{1 - F(t)} = \delta(t), 0 < t < \infty
\end{aligned}$$

Note that,

$$\begin{aligned}
&\int_0^t \frac{F'(x)}{1 - F(x)}dx = \int_0^t \delta(x)dx \\
\Rightarrow &-\ln(1 - F(x))]_0^t = \int_0^t \delta(x)dx \\
\Rightarrow &-\ln(1 - F(t)) + \ln(1 - F(0)) = \int_0^t \delta(x)dx, [F(0) = P(T \leqslant 0) = 0 \text{as} T \text{is a positive RV}] \\
\Rightarrow &1 - F(t) = e^{-\int_0^t \delta(x)dx} \\
\Rightarrow &F(t) = 1 - e^{-\int_0^t \delta(x)dx}, \quad 0 < t < \infty
\end{aligned}$$

In particular: $F(t) = 1 - e^{-\int_0^t \theta dx} = 1 - e^{-\theta t}$

$\therefore T \sim$ Exponential distribution with mean $\frac{1}{\theta}$.

7 The converse of the Lack of Memory Property

If a positive continuous $RV X$ has the lack of memory property,

$$P[X > a + b \mid X > a] = P[X > b] \quad a > 0, b > 0$$

Then, X must have an exponential distribution.

Proof: Let $F(x)$ be the CDF of X.

$$\begin{aligned} & P[X > a + b \mid X > a] = P[X > b] \\ \Rightarrow & P[X > a + b] = P[X > a]P[X > b] \\ \Rightarrow & \{1 - F(a + b)\} = \{1 - F(a)\}\{1 - F(b)\} \end{aligned}$$

[If $g(x + y) = g(x) + g(y) \quad \forall (x, y)$, the only continuous solution of $g(x)$ is

$$\begin{aligned} & g(x) = ax \\ \Rightarrow & \ln\{1 - F(a + b)\} = \ln\{1 - F(a)\} + \ln\{1 - F(b)\}. \\ \Rightarrow & g(a + b) = g(a) + g(b) \quad \text{where } g(x) = \ln\{1 - F(x)\}. \\ \Rightarrow & g(x) = -\theta x, \quad 0 < x < \theta \text{ and } 0 < \theta < \infty \\ \therefore & \ln(1 - F(x)) = -\theta x \\ \Rightarrow & F(x) = 1 - e^{-\theta x} \quad 0 < x < \infty, \theta > 0 \end{aligned}$$

$\therefore X \sim$ Exponential distribution with mean $= \frac{1}{\theta}$.

Remark: The lack of memory property is a unique property of an Exponential distribution among all possible continuous RV. This is a characterization of an exponential distribution in the class of all positive continuous distribution.